Section 5 | Preliminary Results and Validation

CIS 480 Capstone • AI4I 2020 Predictive Maintenance • T07 Team Package • 10/29/2026

Evidence scope: publicly available synthetic AI4I 2020 data, team modeling notebook exports, and Tableau workbook draft. Results below are preliminary; they do not authorize operational deployment.

5.1 PROJECT QUESTION AND EVALUATION APPROACH

Can machine-learning models identify potential equipment failures from AI4I 2020 operating measurements, and which model offers a useful balance between detecting failures and limiting false alarms? The team compared Random Forest and Logistic Regression on the cleaned dataset. The cleaned test partition contains 1,995 records, including 66 failures and 1,929 non-failures. We also compare Random Forest performance on the original dataset as a sensitivity check. Confusion matrices are the primary evidence for the decision.

5.2 DECISION-RELEVANT OUTPUT 1: MODEL COMPARISON

Metric (cleaned test set)	Random Forest	Logistic Regression	No-failure baseline
True positives / false negatives	52 / 14	11 / 55	0 / 66
False positives / true negatives	28 / 1,901	4 / 1,925	0 / 1,929
Recall	78.79%	16.67%	0.00%
Precision	65.00%	73.33%	Not defined (0 predicted failures)
Accuracy	97.89%	97.04%	96.69%
False-positive rate	1.45%	0.21%	0.00%

Table 1. Model outcomes on 1,995 cleaned test observations. The no-failure baseline always predicts normal operation. Its high accuracy is driven by the imbalance between normal records and failures; it detects no failures. The baseline is a simple decision reference, not the same as the separately trained balanced Logistic Regression reported in the earlier T06 pipeline.

Interpretation: Random Forest catches 52 of 66 failures, versus 11 for Logistic Regression and none for the no-failure baseline. Logistic Regression raises fewer false alarms (4 versus 28) and has higher precision, but misses 55 failures. For a preliminary failure-screening use case, Random Forest is the stronger candidate, subject to review of the operational costs of missed failures and false alarms.

5.3 DECISION-RELEVANT OUTPUT 2: ORIGINAL-VERSUS-CLEANED RANDOM FOREST

Metric	Original dataset	Cleaned dataset	Observed direction
Recall	77.94%	78.79%	Higher
Precision	58.89%	65.00%	Higher
F1 score	67.09%	71.23%	Higher
Average precision	0.7543	0.8030	Higher
False-positive rate	1.92%	1.45%	Lower

Table 2. Wynn's interpretation of Brett's Random Forest notebook results. The model shows broadly similar failure detection with improved reported precision and fewer false alarms

on the cleaned evaluation. Because the original and cleaned datasets use different training/test splits, these results cannot isolate the causal effect of cleaning; a paired evaluation on a fixed test set would be needed.

5.4 INDEPENDENT VALIDATION AND EXPECTED-VERSUS-ACTUAL RECORD

Independent arithmetic check using the saved confusion-matrix counts: Random Forest recall = TP / (TP + FN) = 52 / (52 + 14) = 52 / 66 = 78.79%. Its false-positive rate = FP / (FP + TN) = 28 / (28 + 1,901) = 28 / 1,929 = 1.45%. Accuracy = (52 + 1,901) / 1,995 = 97.89%. Logistic Regression recall = 11 / (11 + 55) = 16.67%. No-failure baseline accuracy = 1,929 / 1,995 = 96.69%.

Input / question	Expected result	Actual result	Status / evidence
RF recall: TP=52, FN=14	52/66 ≈ 78.79%	78.79%	PASS — RF cleaned notebook PDF
RF false-positive rate: FP=28, TN=1,901	28/1,929 ≈ 1.45%	1.45%	PASS — RF cleaned notebook PDF
LR recall: TP=11, FN=55	11/66 ≈ 16.67%	16.67%	PASS — LR cleaned notebook PDF
Majority-class baseline: 1,929 normal, 66 failures	96.69% accuracy, 0% recall	96.69% / 0%	PASS — independent calculation; T06 baseline CSV
Tableau draft can open without local data path	Portable data connection	Local path in .twb; .twbx not supplied	PENDING — Tableau .twb draft

Table 3. Validation record. PASS means the stated arithmetic agrees with the saved outputs, not that an independent end-to-end notebook rerun was completed. Notebook PDFs and the Tableau draft are included in the evidence package.

5.5 DECISION RULE AND LIMITATIONS

Preliminary screening rule: favor a model that detects substantially more failures than the no-failure baseline without an excessive false-alarm rate. The team's proposed evaluation targets are at least 75% recall and at most 2% false-positive rate; these are provisional project targets, not externally validated safety or industry standards. Random Forest meets both on this saved cleaned test split. Logistic Regression does not meet the recall target.

Supported: on this test partition, Random Forest detects more failures than Logistic Regression and the majority-class baseline; its saved false-positive rate is below 2%. Unknown: how either model performs on actual equipment, under distribution shift, on a truly independent production dataset, or after threshold calibration. The AI4I dataset is synthetic, failures are rare, and feature importance does not establish causation. The observed differences between original and cleaned evaluations do not prove a cleaning effect. No production deployment is recommended based on these results alone.

5.6 TABLEAU VISUALIZATION STATUS AND ACCESSIBILITY

Zach prepared four draft worksheets from the cleaned data: Air Temperature vs. Process Temperature, and Torque vs. Rotational Speed separately for machine Types H, L, and M. These are supplementary views of operating measurements, not direct validation of the classifier. The .twb file is included for inspectability, but it references a local source path; a portable .twbx and any final chart interpretations are to be placed in GitHub. The notebook results and calculations above provide the two primary decision outputs without relying on the Tableau connection.

Accessible text description of the visual evidence: the model confusion matrices summarize actual failures and non-failures against predicted outcomes. The cleaned Random Forest matrix contains 1,901 true negatives, 28 false positives, 14 false negatives, and 52 true positives. The cleaned Logistic Regression matrix contains 1,925 true negatives, 4 false positives, 55 false negatives, and 11 true positives. These counts are also presented in Table 1 so the information does not depend on images.

5.7 CURRENT WORK PLAN AND CONTRIBUTION EVIDENCE

Owner	Task / status	Dependency	Acceptance criterion	Contribution evidence / effect
Brett	Model notebooks and results — COMPLETE; confirmed current outputs	Cleaned AI4I dataset	Inspectible RF and LR outputs	Notebook exports and GitHub commits; establishes model evidence
Wynn	Interpret performance and cleaning comparison — COMPLETE	Brett's reported metrics	Accurate metric comparisons and caveats	Written original/cleaned comparison incorporated into §5.3
Zach	Tableau draft worksheets; packaged upload pending	Cleaned dataset and portable data connection	Four readable charts; portable .twbx	CIS_480_Semi_Draft.twb; supplementary operating-condition analysis
Keeshaun	Write Section 5, independently verify metrics, assemble evidence — COMPLETE;	Brett's results, Wynn's interpretation, Zach's draft	Complete accessible package and checked calculations	This report, validation record, and evidence package

Table 4. Current ownership and acceptance record.

5.8 EVIDENCE, SOURCES, AND AI-USE DISCLOSURE

Team repository: https://github.com/37216110/cis480_capstone

Modeling notebook directory

https://github.com/37216110/cis480_capstone/tree/main/notebooks/modeling

Brett's original modeling commit

https://github.com/37216110/cis480_capstone/commit/3b0f642

Brett's leakage-check update

https://github.com/37216110/cis480_capstone/commit/78c34cb

Brett's original/cleaned rerun update

https://github.com/37216110/cis480_capstone/commit/37ebee0

Supporting Evidence

The submission includes Brett's Random Forest and Logistic Regression results for both the original and cleaned datasets, Zach's preliminary Tableau visualizations, and the baseline results from T06. These files support the model comparisons, validation calculations, and findings presented in Section 5.

AI-use disclosure: ChatGPT assisted with report organization and formatting. Brett's model outputs and Wynn's written comparison were used as the substantive source evidence.